

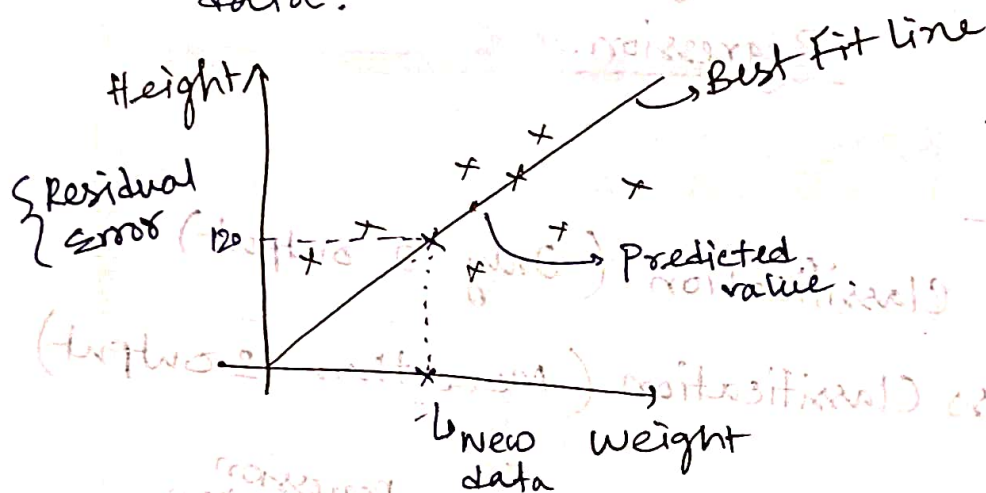
① Linear Regression In Machine Learning Algo:-

Agenda :-

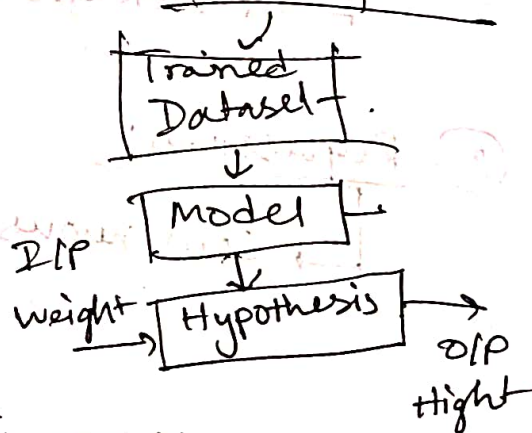
- ① What problem we are solving
- ② Geometric Intuition
- ③ Mathematical Intuition.

① What problem we are solving?

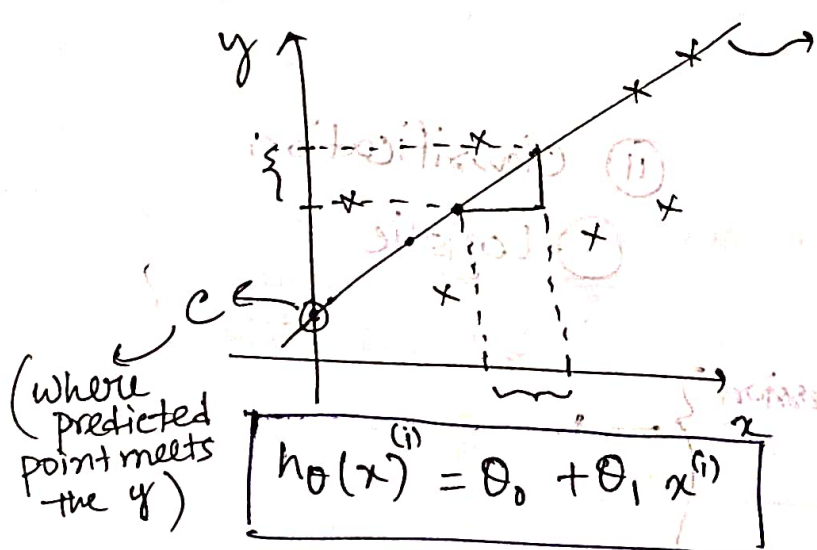
→ Predict a continuous value based on input data.



Weight	Height
73	150
63	155
78	170
82	165



Maths :-



$$\hat{y} = mx + c$$

m = slope or coefficient
 c = Intercept
 $\hat{y}$ = predicted output

$$h_0(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$$

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

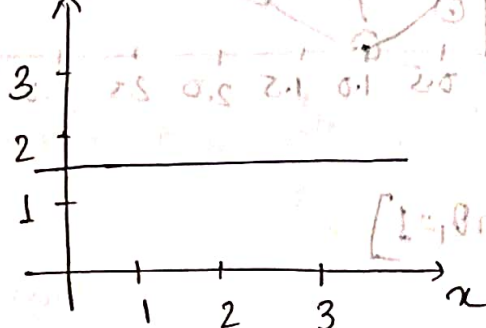
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Hypothesis :- y is a linear function of x

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 \rightarrow \text{for best fit.}$$

[where θ_0 = Intercept
 θ_1 = Coefficient]

Q.1:-



$$\theta_0 = 1.5$$

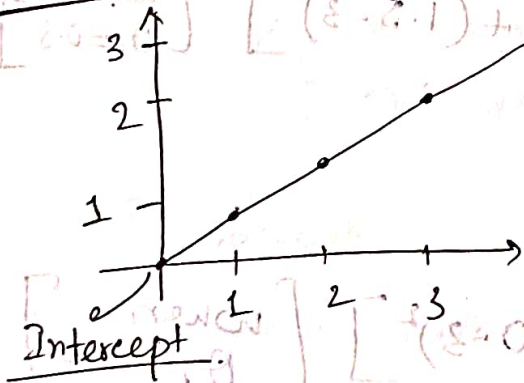
$$\theta_1 = 0$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1$$

$$= 1.5 + 0 \cdot x_1 = 1.5$$

$$h_{\theta}(x) = 1.5$$

Q.2:-



$$\theta_0 = 0$$

$$\theta_1 = 0.5$$

for $x_1 = 1$

$$[h_{\theta}(x) = 0 + (0.5) \times 1 = 0.5]$$

$$\text{for } x_1 = 2 [h_{\theta}(x) = 0 + (0.5) \times 2 = 1]$$

$$\text{for } x_1 = 3 [h_{\theta}(x) = 0 + (0.5) \times 3 = 1.5]$$

Cost function :-

[minimize
 θ_0, θ_1]

$$\sum_{i=1}^m \frac{1}{2m} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

[m $\rightarrow$ no. of datapoints]

changing

Cost function :-

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$\rightarrow$ Squared error function.

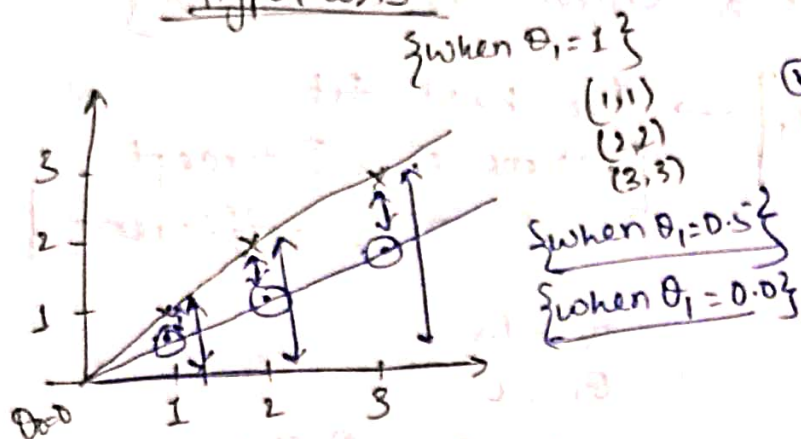
Hypothesis

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1$$

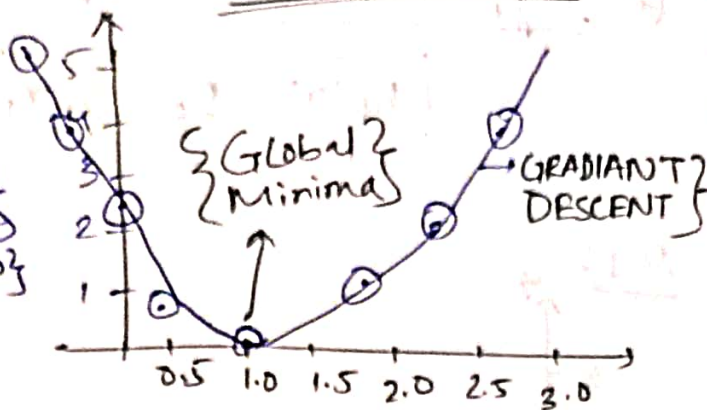
Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Hypothesis



Cost function



$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 \quad [\text{when } \theta_1 = 1]$$

$$= \frac{1}{6} [(0)^2 + (0)^2 + (0)^2]$$

$$= 0$$

when $\theta_1 = 0.5$

$$J(\theta_1) = \frac{1}{6} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] \quad [\text{when } \theta_1 = 0.5]$$

$$= \frac{1}{6} (3.5)$$

$$\approx 0.58$$

when $\theta_1 = 0.0$

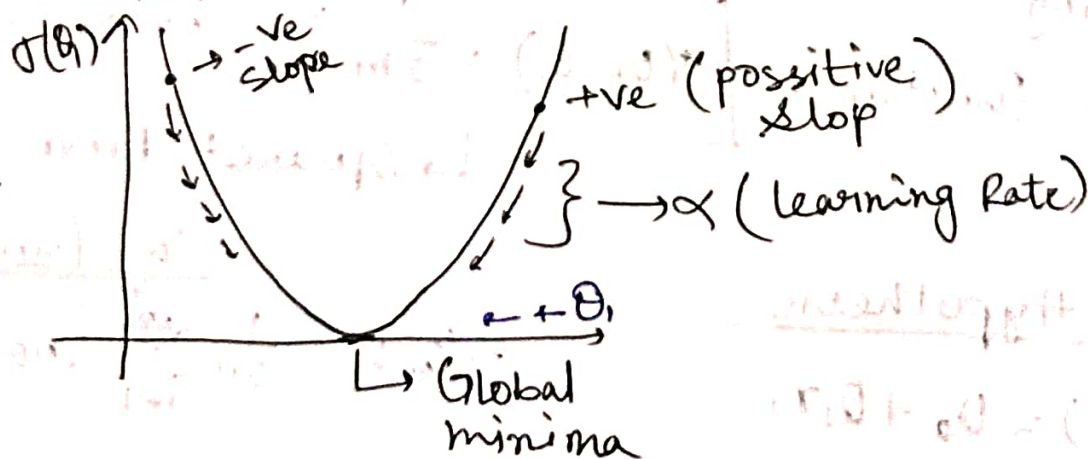
$$J(\theta_1) = \frac{1}{6} [(0-1)^2 + (0-2)^2 + (0-3)^2] \quad [\text{when } \theta_1 = 0.0]$$

$$= \frac{1}{6} [1 + 4 + 9]$$

$$= \frac{1}{6} \times 14$$

$$\approx 2.3$$

⑧ Repeat Convergence Theorem:-



Repeat Convergence Theorem:-

$$\theta_j := \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} (J(\theta_0, \theta_1)) \right] \quad \left[\alpha = \text{Learning Rate} \right]$$

$$\begin{aligned} \theta_1 &:= \theta_1 - \alpha (+ve) \quad \left. \begin{array}{l} \text{Decrease} \\ \text{stop} \end{array} \right\} \\ \theta_1 &:= \theta_1 - \alpha (-ve) \quad \left. \begin{array}{l} \text{Increment} \end{array} \right\} \end{aligned}$$

Learning Rate (α) :-

The learning rate (α) determines how big a step we take in each iteration to minimize the loss.

Real Life ex:-

Descending a mountain blindfold and the learning rate controls the ~~rate~~ length of the step.

(i) small step: safe but slow.

(ii) Big step: fast but might fall or overshoot.

Outline:-

- ① Start with θ_0 and θ_1
- ② Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we reach near global minima.
- ③ Convergence Theorem.

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} (J(\theta_0, \theta_1)) \quad (j = 0 \text{ and } 1)$$

Mean Squared Error, Mean Absolute Error and RMSE :-

$$\textcircled{1} \underline{\text{MSF}} = \boxed{J(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

where
 n = no. of data points
 y_i = Actual value
 $\hat{y}_i$ = Predicted value

(a) Where is MSE used?

-
- Regression problems (Like Linear Regression)
 - Model evaluation during training.
 - Loss function in Optimization algorithm.

Smaller MSE = Better Model

Advantages

- (a) Differentiable
- (b) It has one local minima and one global minima.

Disadvantages

- (a) Not Robust to outliers
- (b) It changes its unit.

Example:-

If you have actual values: $[3, 5, 2]$ and predicted values: $[2, 5, 4]$

$$\rightarrow \text{MSF} = \frac{1}{3} [(3-2)^2 + (5-5)^2 + (2-4)^2]$$

$$= \frac{1}{3} [1 + 0 + 4]$$

$$= \frac{5}{3}$$

$$\approx 1.67$$

② Mean Absolute Error (MAE):-

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where,
 n = no. of data points
 y_i = Actual value
 $\hat{y}_i$ = Predicted value
 $| \cdot |$ = Absolute value
(ignores negative sign)

① characteristics:-

- (a) Does not square the errors (unlike MSE), so it is less sensitive to outliers.
- (b) Gives equal weight to all errors.
- (c) More robust when large errors are not more critical than small ones.

Advantages

- (a) It is robust to outliers.
- (b) It will be same unit.

Disadvantages

- (a) Convergence usually takes more time.
- (b) Optimization is a complex process.

⑧ Example:-

Actual : [3, 5, 2]

Predicted : [2, 5, 4]

$$\rightarrow MAE = \frac{1}{3} [|3-2| + |5-5| + |2-4|]$$

$$= \frac{1}{3} (1+0+2) = 1$$

③ RMSE :- (Root Mean Square Error)

$$RMSE = \sqrt{MSE}$$